

InterRings: Towards Understanding Design Micro-games to Fit Daily Work Routine

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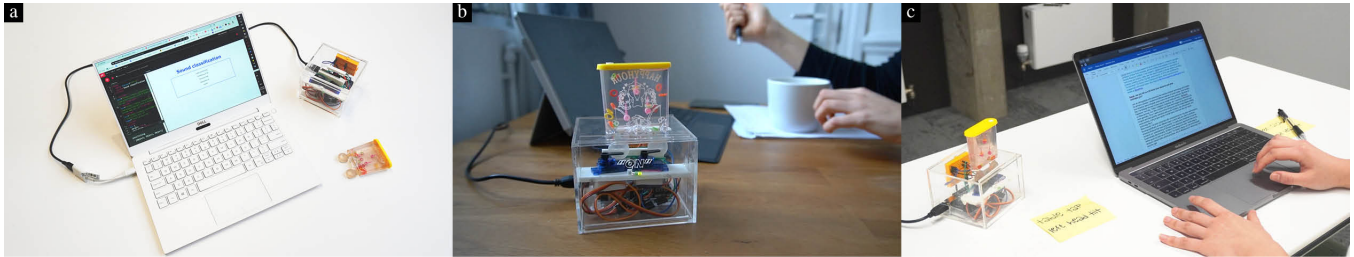


Figure 1: (a) Assembly of InterRings. (b) The sounds of pen clicking (left servo) and table tapping (right servo) activate two servos respectively to press airbags, thus to stack rings on the peg. (c) A participant is undertaking a time-log observational study.

ABSTRACT

Prior game researchers mostly focused on designing the "macro" context which requires a decent proportion of time to engage and play, from which players may benefit. However, the demanding time and effort restrict playability during working hours. To our knowledge, there is little research exploring game design in a manner that enables momentary detachment from primary work for a mind refreshment and effortlessly resumes the suspended work after playing. In this paper, we propose a novel micro-game concept that enables the gamification of micro-breaks within working hours. To examine our concept, we adapted a conventional water-ring game into an interactive prototype, InterRings, which empowers users to play with handy objects. Our mixed observational studies and interviews revealed high acceptability and feasibility of the micro-game concept. Conclusively, three design guidelines are summarised for future development of micro-games.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design theory, concepts and paradigms.**

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KEYWORDS

Micro-games; Ubiquitous Computing; Multi-tasking; Workplace Well-being

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1 INTRODUCTION

In recent game design development, the tendency to apply games to a variety of areas, such as physical fitness [8, 15], alternative therapy for mental health [5, 24], and education [27, 36, 42], is prominent. Current game research, however, mainly focused on recovery activities outside of workplace during leisure times, little attention has been paid to gamifying short-term breaks during working hours [23]. Meanwhile, taking regular tiny breaks benefits workers regarding occupational well-being [23, 26, 41], which shows a great potential for researchers to explore. We therefore present InterRings, a novel micro-game that is leveraged to refresh office workers within working hours. It aims at promoting temporal detachment from studying and working while enabling swift resumption of work, and consequentially helps users develop a sustainable work routine. Overall, this paper contributes to the following areas in terms of micro-games:

- We reviewed previous game taxonomy and then categorized existing games into four clusters based on two dimensions: time and effort required. Based on these criteria, we extended the definition of micro-game.
- We described the fabrication process of our micro-game prototype, InterRings and then introduced how ubiquitous

interfaces could be employed to create a smooth work and play transition.

- We ran a time-log observation study of a small batch of users to evaluate our micro-game concept, which served as the basis of our three guidelines to support future development of micro-games.

2 RELATED WORK

2.1 Micro-breaks

Individuals are occupied by a series of discrete events on workdays and become bored and fatigued during prolonged working hours. This causes a slump of motivation and morale, which further leads to inefficiency in the long run [35]. Taking breaks is beneficial when transiting between work-related events, because it can mitigate the negative effects of sustained stimulus on energy based resources [40]. In the context of work-related stress, playing digital games also shows a positive correlation with work-related stresses recovery [10, 33]. A previous study also released that taking micro-breaks encourages workers to perform more tasks to some extent [35].

Taking micro-breaks, as an energy management strategy, is essential and necessary for occupational well-being. For instance, strategic short breaks preserve mental resources and have been found to increase vitality and well-being, reduce fatigue, and restore attention [23, 26, 41]. Even a tiny 3-30 seconds break can make a huge difference [35]. In particular, low-effort and enjoyable activities that promote detachment from work are especially effective at recovering mental resources and avoiding stress [39].

2.2 Ubiquitous game interface

Ubiquitous computing refers to the system where computers are physically and socially embedded in our natural motion and interactions with surroundings [28]. Such environment emerged trend also appears in game interfaces. Actually, ubiquitous game interfaces have already been applied to game-based learning [27, 36, 42], interactive experience enrichment [29], and traditional game enhancement [12, 34]. Taking the example of Ubiballs [12], ubiquitous computing extends digital game play further into reality. The project exhibits a promising future of games that are not simply to entertain individuals but also potentially to encourage physical fitness [12].

Although ubiquitous interface offers significant insights about physical and social context, even recognising real-time activity [25], the mainstream ubiquitous game interfaces still heavily rely on body tracking and designated accessories. The former offers great flexibility, yet fails to provide precise and tangible feedback that can enhance the game play experience [22, 30]. The latter requires players to possess appropriate input devices in advance, further limiting its applications under ubiquitous gaming scenarios.

2.3 Game classification

In the field of game research, classification of games has constantly been a heated topic [4, 6]. The classification is vital not only because the ability to precisely distinguish and describe games is fundamental for game design process, but also for the fact that it is an important tool in the process of creating or amending formal or semi-formal design tools themselves [13].

Several terms and criteria were attempted to classify games [19], such as style of play [18], mechanics of game play, and player behaviour and experience [21]. Based on literature review and empirical study, Heintz and Law [19] improved the game classification and recommended five genres: mini-game, action, adventure, role-play, and resource. Those genres are defined according to nine attributes: players, I/O, actions, challenges, goals, rewards, settings, perspective, and structure.

Micro-games or mini-games, were commonly described as short, self-contained, naughty, educational, and simply logical [14]. They are widely applied to educational training [14, 18, 32, 37]. Compared with conventional games, micro-games are usually regarded to sit on the opposite [38]. Micro-games usually do not have a complete story line throughout the game. This feature helps players to start the game instantly without having to understand its background and settings. In addition, players can only spend a short period of time to enjoy the game [37, 38]. The game continues unless players want to quit. Besides, micro-games only require simple and repetitive inputs, unlike traditional games which rely on complicated manipulations and have difficulty adjustment mechanisms [18]. Such inputs provide players limited but useful feedback. Think of Running Dinosaur Game [1], a built-in browser game for Google Chrome, as a good example of micro-games. It is well-developed based on an infinite looped story line. Users start the game and control the pixel dinosaur by pressing the Up Arrow key or the Tap key. Comparably, Star Wars: Tales from the Galaxy's Edge [2], as a popular VR adventure game, is considered time-consuming and tricky. Such discrepancy originates from the fact that users have to spend a long time setting up and debugging the device, letting alone an immersive game session is usually long. Meanwhile, players are expected to have a good command of the operations.

3 CONCEPT

3.1 Micro-games

Based on literature review, we characterise micro-game (Fig 2) as a game with simple game dynamics which requires minimal time and effort to play. Micro-games potentially contribute to benefit transition between work and play through gamification during short breaks. Compared to conventional games, micro-games are

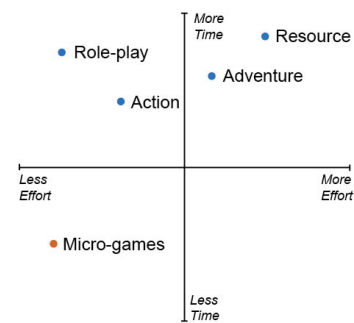


Figure 2: Micro-game is defined as a game that requires minimal effort and time to play.

swiftly developed and iterated, relatively cheaper, enjoyable, and easy to understand [17]. In such scenario, minimal time to play means that the game should have an easy-to-join and easy-to-stop game mechanism. The game should be simple and intuitive to play, thus do not hinder the priority of work. The least effort to play refers to the least switching cost between enjoying playful micro-breaks and resuming previous work. To meet such characteristics, we committed to playing the game without additional input devices such as a dedicated designed game controller. Instead, we proposed to play the game with simple gestures, or with objects that are closely related to the processing work to diminish both physical and cognitive barriers required to play. By merging those elements, we supposed the micro-game concept could be effective to provide workers timely relaxation.



Figure 3: A classical water-ring game which inspired the design of InterRings.

3.2 InterRings

To better understand how stakeholders interact with our micro-games concept in practical working settings, we built a primitive interactive desktop installation, InterRings. Our idea derived from the classical handheld water-ring game (Fig 3) in which a player manipulates water flow inside a plastic tank and aims to stack all colourful rings on the pegs. We selected water-ring game as the foundation for this project for the following reasons. Firstly, its colourful appearance and unpredictability of the water flow increase its playfulness. Combined with simple game dynamics, the game delivers a relaxing and effortless play experience. Secondly, the repetitiveness and slight frustration of the game limit the interaction time, thus making it ideal for micro-breaks. Lastly, the water-ring mechanism auto-saves the current game status. The stacked rings remain on the pegs if no one is playing, which made later resumption of the game possible. InterRings, as a ubiquitous iteration of the classical water-ring game, was embedded with more sustainable features to make it adapt to micro-game characteristics. Instead of pushing the buttons, players can play the game with arbitrary office objects (e.g. pen/glass clicking) or gestures (e.g. tapping/stomping). A sound-based machine learning model would analyse the input gestures to activate the game. For this prototype, we only made use of two inputs, pen clicking and table tapping, for our sound recognition model.

3.3 Implementation

The functional prototype contains two parts: the machine learning algorithm for sound recognition and interactive mechanism to press the buttons. The mechanism consists of Arduino UNO and two attached servos (SG92S) which are connected to a 13 inch laptop (XPS 13 9370). Sounds are firstly captured by the laptop microphone and then analysed by Google's Teachable Machine [3], an open source machine learning platform. These are then assigned to either left or right servo according to the pre-defined classification. For our initial prototype, the right servo was triggered by the sound of pen clicking, while the left servo was triggered by the sound of table tapping. We then employed serial communication software to send signals to Arduino which controls servos and thus to control the water flow. To accommodate these components, we implemented 3D modelling software to locate component position. We then laser cut a transparent acrylic board to house respective components.

4 METHOD

Five participants (2 male and 3 female) from a local business consulting institution were recruited. To understand and evaluate participants interaction cycle, subjects were provided with the interactive prototype within 30 minutes working cycle, and their work and play behaviours were recorded for further analysis. The research was arranged onsite at the consulting firm where staffs work in cubicles in a shared office environment. We first sought informed consent and explained the working principle of our prototype to let users get accustomed to it. Participants were informed to interact with the InterRings prototype at their own pace. We kept a log of what events motivated them to play, when they play, and the duration of play. Afterwards, subjects were invited to participate in a semi-structured interview to help researchers understand their motivations, barriers, and comments on our micro-game concept.

5 RESULTS AND DISCUSSION

5.1 General feedback

Positive feedback was presented about the overall game play process of InterRings, as the game was generally described as a playful, engaging, low-effort, and mind refreshing practice within routine workflows. It is especially effective at facilitating temporal detachment from work. Subjects intended to insert InterRings into their workflows in an intermittent fashion. They frequently switched to play, had a quick treatment, and then swiftly resumed their work. Figure 5 shows participants spent an average of 3:44 min (SD = 1:54 min) to engage with the interactive prototype during the 30 minutes slot, and the average playtime per session was 1:29 min (SD = 0:53 min). The easy-to-start and the easy-to-quit mechanism was also praised by subjects because of its time constraint and sustainability within workspace. Cited from P2, our InterRings prototype *"puts kinds of playful lens on that workspace"*. On the other hand, subjects expressed concerns that the intuitive game mechanism may lead to addiction. Quoted from P3, *"Something started being very easy to play, fun enough, but really addictive to keep going forever."* Aside from that, subjects expressed concerns regarding the frustration experienced by the unpredictability of ring movement, although

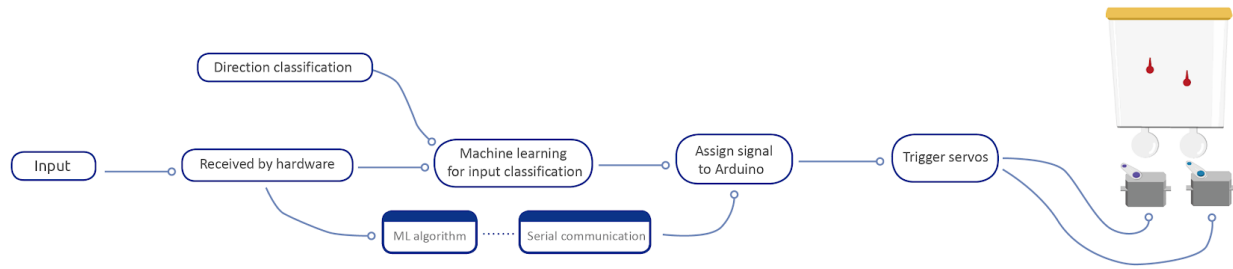


Figure 4: Overview of the InterRings interactive prototype.

they agreed that such unpredictability in the game design helped them get back to work in time. Lastly, subjects mentioned their worries about potentially negative influence on other co-workers in the shared environment as InterRings is a "sound-based" game.

5.2 Fragmented nature of play

As was described by the time-log of game play, subjects showed a strong tendency to insert micro-games into their workflows by constantly switching between artefacts of different purposes for play and work. This is in line with the previous research [7, 16] demonstrating the fragmented nature of information work. Subjects are engaged with a series of connected short-term events in their workflows. Quotes from P2 who confirmed such notion, *"Sometimes I played the game as I hit a mental block, and I quickly resumed my work after having a fresh mind."*

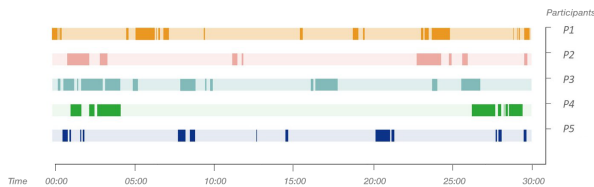


Figure 5: Time log spent on game play in darker colours and summary written work in light colours.

5.3 Motivations of play

Our observation also demonstrated that interruption and task accomplishment are the two contributing factors to trigger game play behaviours. We recorded 61 repetitions of micro-game play in total. 29 times were induced by interruptions such as chatting with colleagues and answering phone calls; 23 of them were activated by completing current events such as finishing writing emails and documents; only 9 game playing behaviours did not belong to the categories mentioned above. This means micro-games could be facilitated as both lubricant and treatment for work routines. The gamification of disruptive interruptions diminished its negative impacts, leading to smoother workflows. Furthermore, micro-games could also served as a timely reward for task accomplishments by the participants.

5.4 Implications

Based on the initial research insights, we summarise three fundamental design guidelines to help design the future micro-game concept.

- Introducing intuitive yet slightly "fricative" game mechanisms to ensure easy game entry and quit. The instinctive game play rules could greatly reduce the learning threshold, therefore encompass wider stakeholders such as amateur players. At the same time, it grants stakeholders to play in an effortless manner that utilizes the least cognitive and physical effort. To limit game interaction time and avoid addiction, we recommend introducing certain levels of friction to intentionally break interaction flows [11, 20]. It is worth noting that those designated difficulties should not counteract an appealing game experience.
- Gamification only at the designated occasion. This ensures the game does not hinder primary work. The game itself should serve as lubricant but not distracting from the workflow [35]. This requires researches to identify potential interactions and methods in advance. For instance, when will the task and rest transition happen? What artefacts may be involved in related tasks? The merge of ubiquitous interfaces, that manipulate everyday objects as game inputs, may well satisfy the micro-game context.
- Compliance to work site restrictions. This requires the resulting games to be non-obtrusive to all people in the shared workplace including players and non-players. This in return calls for sparingly use of visual, auditory, and tactile feedback.

5.5 Limitations

Despite the fact that our work could be a starting point to design micro-games, the study itself clearly has several limitations. Firstly, the external validity of our research was confined by the limited participant scale and was constrained by only one field site at a consulting company. This means specific factors and behaviours may affect the results of this study. Secondly, the method of "shadowing" techniques may not truly reflect the genuine interactions between subjects and InterRings. The presence of researchers themselves may influence the daily workflows [9, 31]. Besides, our research was initialized directly after the induction, so that the long-term effects

of inserting micro-games into workflows remain unknown. It is also unexplored whether participants may change their behaviours to conform to the newly introduced micro-games strategy.

6 CONCLUSION

In this paper, we present an innovative micro-game concept which aims to deliver delightful yet detachable game experience within micro-breaks in fragmented workflows. We defined two key characters of micro-game, which are minimal time and effort. To examine our concept and inform future development of micro-games, we adapted a classic handheld water-ring game and proposed a novel game interface that leverages the ubiquity of everyday work-related movements and objects. We thus expect to empower effortless and seamless switch between work and play. The mixed observational and semi-structured interview study demonstrated promising feasibility and high acceptability of our micro-game concept. Further, we summarise three design guidelines for micro-game design and we hope to inspire more explorations on creating game experiences that are adoptable to scenarios which are conventionally not regarded as suitable for game play.

7 FUTURE WORK

In the future, we would like to extend our micro-game concept to more working environments and to examine its potential applications with stakeholders with diverse job categories. Furthermore, we plan to implement focused studies to understand the rationale behind micro-game interactions. In addition, we schedule to conduct long-term study to understand how micro-game strategy may alert unhealthy working behaviours, and if introducing micro-game strategy may be rewarding to overall work efficiency.

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REFERENCES

- [1] [n.d.]. Running dinosaur Game - Chrome Web Store. <https://chrome.google.com/webstore/detail/running-dinosaur-game/nihmpmddbbbfadmfpmjhogegbjd>. (Accessed on 02/20/2021).
- [2] [n.d.]. Star Wars: Tales from the Galaxy's Edge. <https://www.ocus.com/experiences/quest/3484270071659289>. (Accessed on 02/20/2021).
- [3] [n.d.]. Teachable Machine. <https://teachablemachine.withgoogle.com/>. (Accessed on 01/02/2021).
- [4] Espen Aarseth, Solveig Marie Smedstad, and Lise Sunnanå. 2003. A multidimensional typology of games.. In *DiGRA Conference*.
- [5] Vasundhara Agrawal, Varnika Naik, Mayuri Duggirala, and Sandeep Athavale. 2020. Color Me: A Game based on Art Therapy for Mental Health. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play*. 158–162.
- [6] Thomas H Apperley. 2006. Genre and game studies: Toward a critical approach to video game genres. *Simulation & Gaming* 37, 1 (2006), 6–23.
- [7] Liam Bannon, Allen Cypher, Steven Greenspan, and Melissa L Monty. 1983. Evaluation and analysis of users' activity organization. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. 54–57.
- [8] Damon Brown. 2006. Playing to win: Video games and the fight against obesity. *Journal of the American Dietetic Association* 106, 2 (2006), 188–189.
- [9] Paul Ed Cairns and Anna L Cox. 2008. *Research methods for human-computer interaction*. Cambridge University Press.
- [10] Emily Collins and Anna L Cox. 2014. Switch on to games: Can digital games aid post-work recovery? *International Journal of Human-Computer Studies* 72, 8-9 (2014), 654–662.
- [11] Anna L Cox, Sandy JJ Gould, Marta E Cecchinato, Ioanna Iacovides, and Ian Renfree. 2016. Design frictions for mindful interactions: The case for microboundaries. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 1389–1397.
- [12] Douglas Easterly and Angela Blachnitzky. 2009. UbiBall: A Ubiquitous Computing Game for Children. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/1621841.1621850>
- [13] Christian Elverdam and Espen Aarseth. 2007. Game Classification and Game Design: Construction Through Critical Analysis. *Games and Culture* 2, 1 (2007), 3–22. <https://doi.org/10.1177/1555412006286892> arXiv:<https://doi.org/10.1177/1555412006286892>
- [14] Alex Frazer, David Argles, and Gary Wills. 2007. Is less actually more? The usefulness of educational mini-games. In *Seventh IEEE International Conference on Advanced Learning Technologies (ICALT 2007)*. IEEE, 533–537.
- [15] Tim Gabbett, David Jenkins, and Bruce Abernethy. 2009. Game-based training for improving skill and physical fitness in team sport athletes. *International Journal of Sports Science & Coaching* 4, 2 (2009), 273–283.
- [16] Victor M González and Gloria Mark. 2004. "Constant, constant, multi-tasking craziness" managing multiple working spheres. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 113–120.
- [17] Lindsay Grace, G Tanner Jackson, Christopher Totten, Julian Parker, and Joyce Rice. 2015. Designing microgames for Assessment: A case study in rapid prototype iteration. In *Proceedings of the 12th International Conference on Advances in Computer Entertainment Technology*. 1–4.
- [18] Lindsay D Grace. 2012. Making and analyzing games: not as art, but as literature. In *Proceeding of the 16th International Academic MindTrek Conference*. 237–240.
- [19] Stephanie Heintz and Effie Lai-Chong Law. 2015. The Game Genre Map: A Revised Game Classification. In *Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play (London, United Kingdom) (CHI PLAY '15)*. Association for Computing Machinery, New York, NY, USA, 175–184. <https://doi.org/10.1145/2793107.2793123>
- [20] Ioanna Iacovides, Anna L Cox, Patrick McAndrew, James Aczel, and Eileen Scanlon. 2015. Game-play breakdowns and breakthroughs: exploring the relationship between action, understanding, and involvement. *Human-computer interaction* 30, 3-4 (2015), 202–231.
- [21] Klaus P Jantke. 2006. Games that do not exist communication design beyond the current limits. In *Proceedings of the 24th annual ACM international conference on Design of communication*. 35–42.
- [22] Dagmar Kern, Mark Stringer, Geraldine Fitzpatrick, and Albrecht Schmidt. 2006. Curbball—a prototype tangible game for inter-generational play. In *15th IEEE International Workshops on Enabling Technologies: Infrastructure for Collaborative Enterprises (WETICE'06)*. IEEE, 412–418.
- [23] Sooyeol Kim, YoungAh Park, and Qikun Niu. 2017. Micro-break activities at work to recover from daily work demands. *Journal of Organizational Behavior* 38, 1 (2017), 28–44.
- [24] Simone Kühn, Tobias Gleich, Robert C Lorenz, Ulman Lindenberger, and Jürgen Gallinat. 2014. Playing Super Mario induces structural brain plasticity: gray matter changes resulting from training with a commercial video game. *Molecular psychiatry* 19, 2 (2014), 265–271.
- [25] Gierad Laput, Karan Ahuja, Mayank Goel, and Chris Harrison. 2018. Ubicoustics: Plug-and-Play Acoustic Activity Recognition. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3242587.3242609>
- [26] Kate E. Lee, Kathryn J.H. Williams, Leisa D. Sargent, Nicholas S.G. Williams, and Katherine A. Johnson. 2015. 40-second green roof views sustain attention: The role of micro-breaks in attention restoration. *Journal of Environmental Psychology* 42 (2015), 182 – 189. <https://doi.org/10.1016/j.jenvp.2015.04.003>
- [27] Tsung-Yu Liu and Yu-Ling Chu. 2010. Using ubiquitous games in an English listening and speaking course: Impact on learning outcomes and motivation. *Computers & Education* 55, 2 (2010), 630 – 643. <https://doi.org/10.1016/j.compedu.2010.02.023>
- [28] Kalle Lyytinen and Youngjin Yoo. 2002. Ubiquitous computing. *Commun. ACM* 45, 12 (2002), 63–96.
- [29] Tony Manninen. 2002. Contextual virtual interaction as part of ubiquitous game design and development. *Personal and Ubiquitous Computing* 6, 5-6 (2002), 390–406.
- [30] Javier Marco, Eva Cerezo, and Sandra Baldassarri. 2009. Evaluating a tangible game video console for kids. In *IFIP Conference on Human-Computer Interaction*. Springer, 141–144.
- [31] Henry Mintzberg. 2019. Structured observation as a method to study managerial work. *Journal of* (2019).
- [32] Chris T Panagiotakopoulos. 2011. Applying a Conceptual Mini Game for Supporting Simple Mathematical Calculation Skills: Students' Perceptions and Considerations. *World Journal of Education* 1, 1 (2011), 3–14.
- [33] Leonard Reinecke. 2009. Games at work: The recreational use of computer games during working hours. *CyberPsychology & Behavior* 12, 4 (2009), 461–465.
- [34] Kay Römer and Svetlana Domnitcheva. 2002. Smart playing cards: A ubiquitous computing game. *Personal and Ubiquitous Computing* 6, 5-6 (2002), 371–377.

- [35] Jeffrey Rzeszutarski, Ed Chi, Praveen Paritosh, and Peng Dai. 2013. Inserting micro-breaks into crowdsourcing workflows. In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, Vol. 1.
- [36] José-Manuel Sáez-López Sáez-López, M Luisa Sevillano-García Sevillano-García, M Ángeles Pascual-Sevillano Pascual-Sevillano, José-Manuel Sáez-López, María Luisa Sevillano-García-García, and María de los Ángeles Pascual-Sevillano. 2019. Application of the ubiquitous game with augmented reality in Primary Education. *Comunicar. Media Education Research Journal* 27, 2 (2019).
- [37] Peter A Smith and Alicia Sanchez. 2010. Mini-games with major impacts. In *Serious game design and development: Technologies for training and learning*. IGI Global, 1–12.
- [38] Stan W. Smith. 2010. An experiment in bibliographic mark-up: Parsing metadata for XML export. In *Proceedings of the 3rd. annual workshop on Librarians and Computers (LAC '10, Vol. 3)*, Reginald N. Smythe and Alexander Noble (Eds.). Paparazzi Press, Milan Italy, 422–431. <https://doi.org/99.9999/woot07-S422>
- [39] John P Trougakos and Ivona Hideg. 2009. Momentary work recovery: The role of within-day work breaks. In *Current perspectives on job-stress recovery*. Emerald Group Publishing Limited.
- [40] Matthew Wilkinson and Caitlin Demsky. 2016. *Micro Breaks*.
- [41] Hannes Zacher, Holly A Brailsford, and Stacey L Parker. 2014. Micro-breaks matter: A diary study on the effects of energy management strategies on occupational well-being. *Journal of Vocational Behavior* 85, 3 (2014), 287–297.
- [42] X. Zhai, Y. Dong, and J. Yuan. 2018. Investigating Learners' Technology Engagement - A Perspective From Ubiquitous Game-Based Learning in Smart Campus. *IEEE Access* 6 (2018), 10279–10287. <https://doi.org/10.1109/ACCESS.2018.2805758>